math448_project

Samantha Bell

2025-09-16

1. reading data into R

```
library(readxl)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(MASS)
```

```
##
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:dplyr':
##
##     select
```

```
library(pROC)
```

```
## Warning: package 'pROC' was built under R version 4.5.2
```

```
## Type 'citation("pROC")' for a citation.
```

```
##
## Attaching package: 'pROC'
```

```
## The following objects are masked from 'package:stats':
##
##     cov, smooth, var
```

```
pcos = read.csv("C:/Users/theresa/Downloads/pcos_dataset.csv")
```

```
# transforming variables
pcos = pcos |>
  mutate(PCOS_Diagnosis = factor(PCOS_Diagnosis)) |>
  mutate(Menstrual_Irregularity = as.numeric(factor(Menstrual_Irregularity))) |>
  mutate(Age = as.numeric(Age)) |>
  mutate(Antral_Follicle_Count = as.numeric(Antral_Follicle_Count))
```

2. processing data

checking for NA's

```
sum(is.na(pcos))
```

```
## [1] 0
```

range of variables

```
t(sapply(pcos[,1:5], range))
```

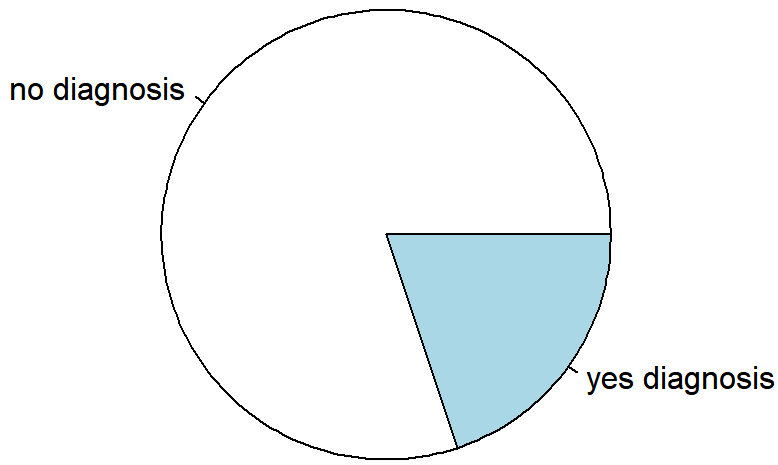
```
##                          [,1] [,2]
## Age                      18.0 45.0
## BMI                      18.1 35.0
## Menstrual_Irregularity    1.0  2.0
## Testosterone_Level.ng.dL. 20.0 99.8
## Antral_Follicle_Count     5.0 29.0
```

checking for balance

```
t = table(pcos$PCOS_Diagnosis)
t
```

```
##
##   0   1
## 801 199
```

```
pie(c(801,199), labels = c('no diagnosis', 'yes diagnosis'))
```



The data I will be using is compiled of 1000 patients and five variables which are typically associated with PCOS diagnosis and its risks. These variables are age, BMI, menstrual irregularity, testosterone level, and number of antral follicles. Finally, the response variable is whether a patient has been diagnosed with PCOS or not, making it a classification data set.

After running a few tests, I see that I have no missing data throughout the 1000 entries and 6 variables. I do, however, have unbalanced data, 80.1% with no diagnosis and 19.9% with a diagnosis. Having such a large unbalance in data can lead to biased outcomes when trying to model the data.

balancing data

```
library(ROSE)
```

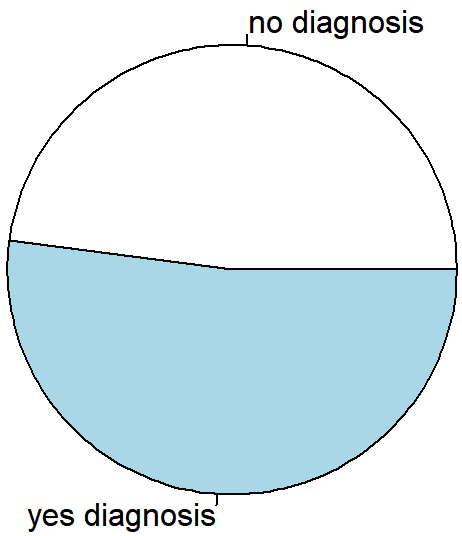
```
## Warning: package 'ROSE' was built under R version 4.5.2
```

```
## Loaded ROSE 0.0-4
```

```
set.seed(1)
pcos = ovun.sample(PCOS_Diagnosis ~ ., data=pcos, method="under")$data
t = table(pcos$PCOS_Diagnosis)
t
```

```
##
##   0   1
## 183 199
```

```
pie(c(183,199), labels = c('no diagnosis', 'yes diagnosis'))
```



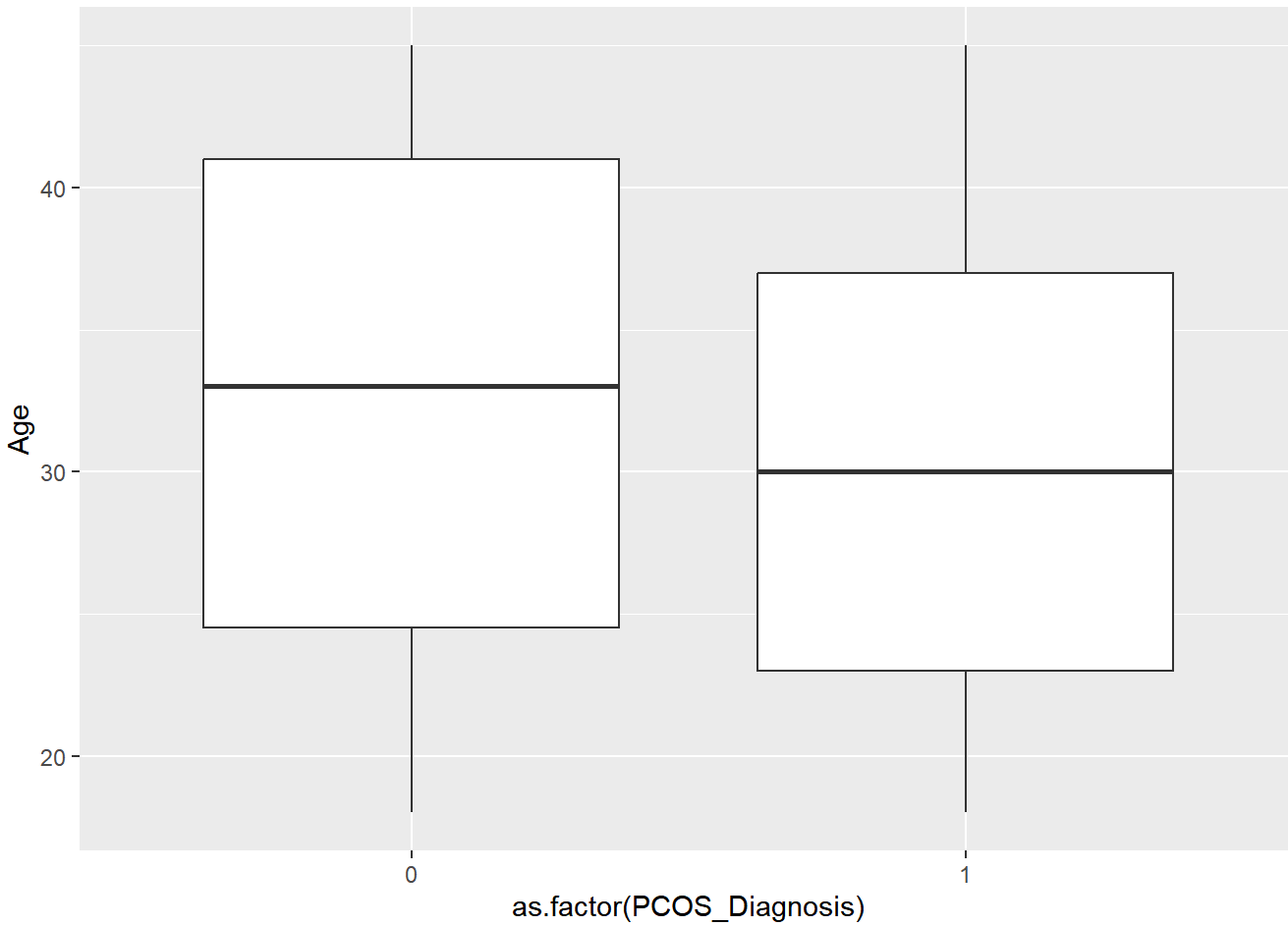

```
t(sapply(pcos[,1:5], range))
```

```
##                          [,1] [,2]
## Age                      18.0 45.0
## BMI                      18.2 35.0
## Menstrual_Irregularity    1.0  2.0
## Testosterone_Level.ng.dL. 20.1 99.6
## Antral_Follicle_Count     5.0 29.0
```

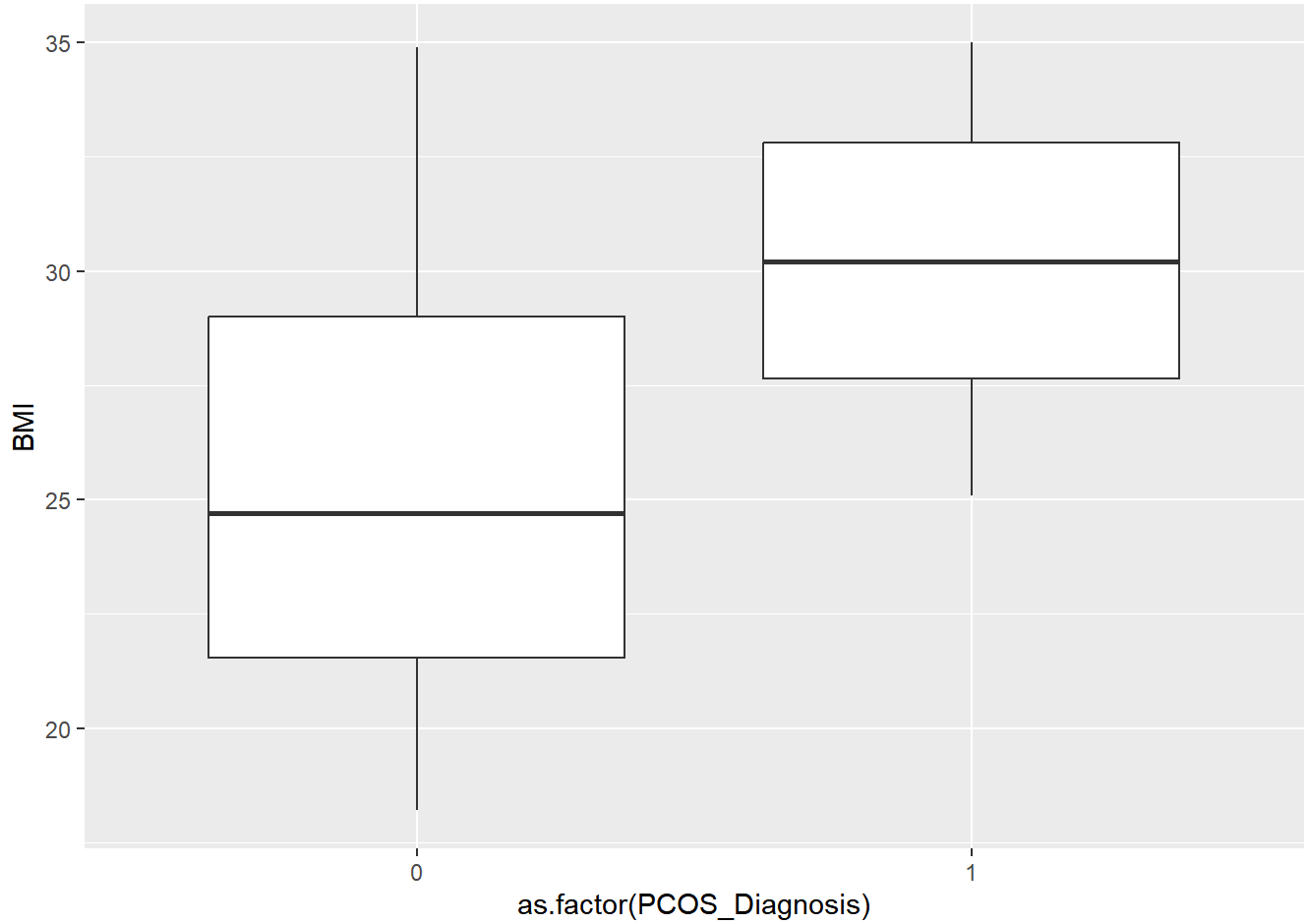
Now I have a more balanced data with only 382 entries, 183 with no diagnosis and 199 with a diagnosis. Also, the range of the cut-down data is the same as the original. So, this data will be a great sample that does not lead to bias.

3. Descriptive Analysis

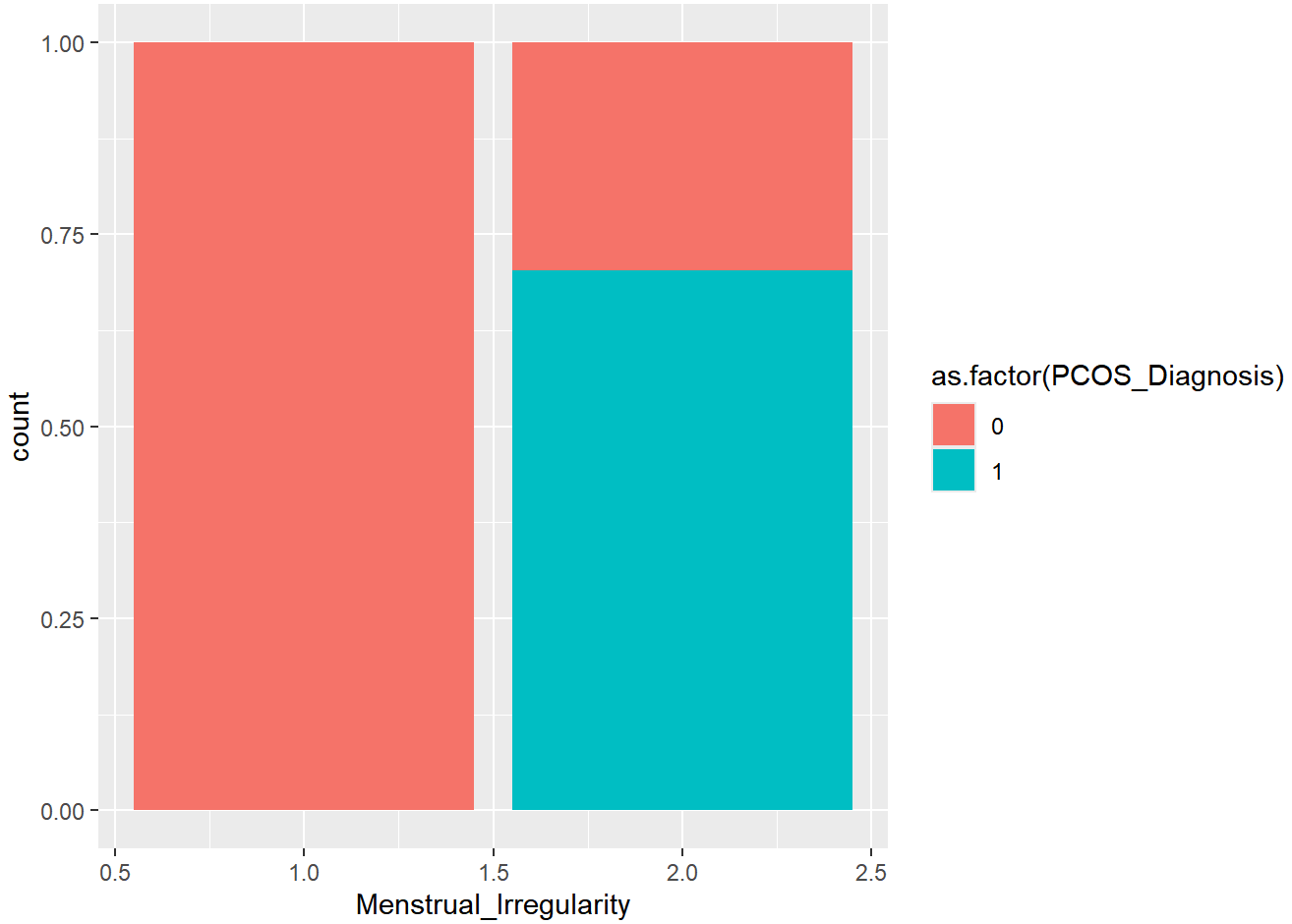
```
library(ggplot2)

ggplot(pcos, aes(x= as.factor(PCOS_Diagnosis), y = Age)) +
  geom_boxplot()
```



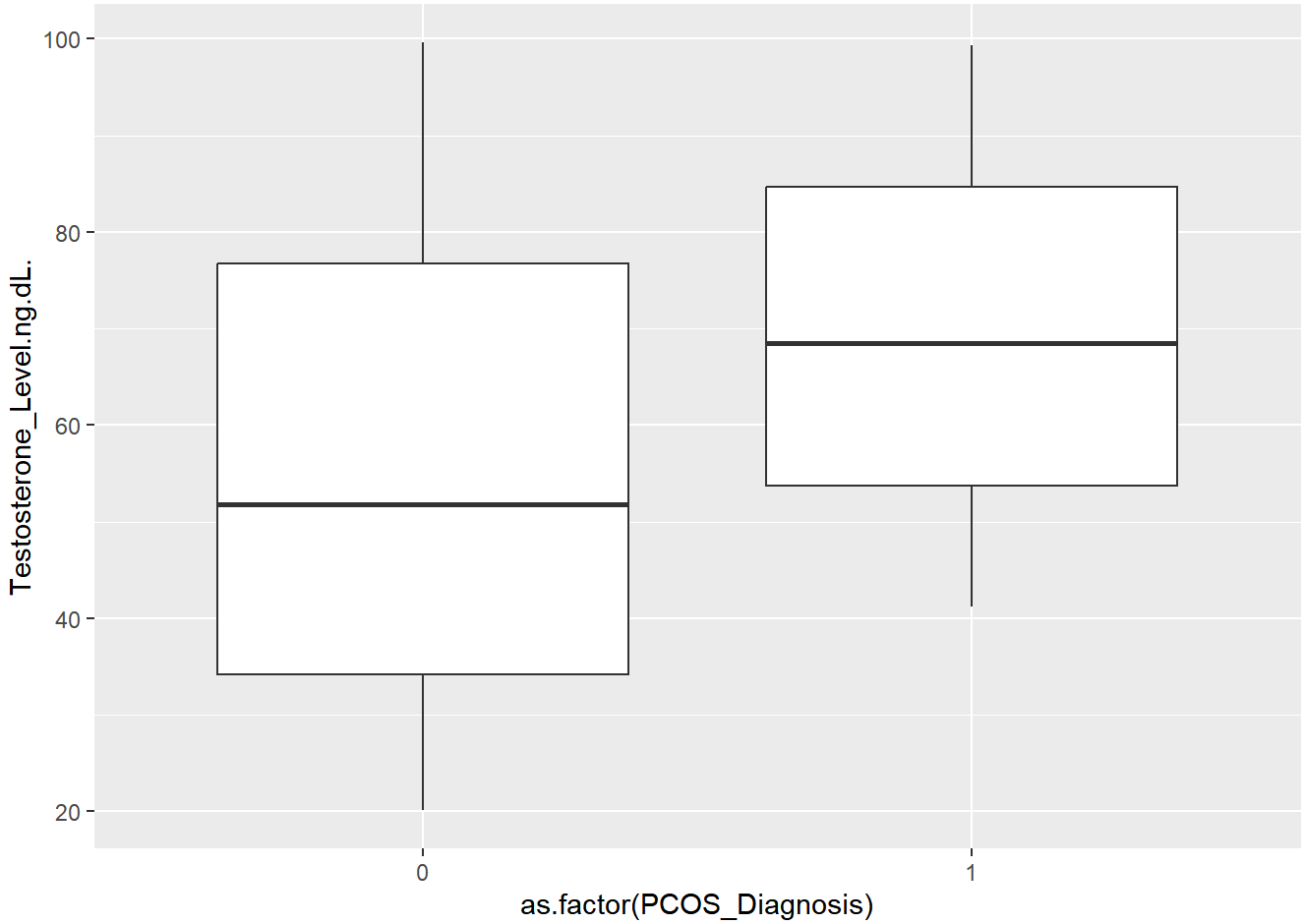

```
ggplot(pcos, aes(x= as.factor(PCOS_Diagnosis), y = BMI)) +
  geom_boxplot()
```



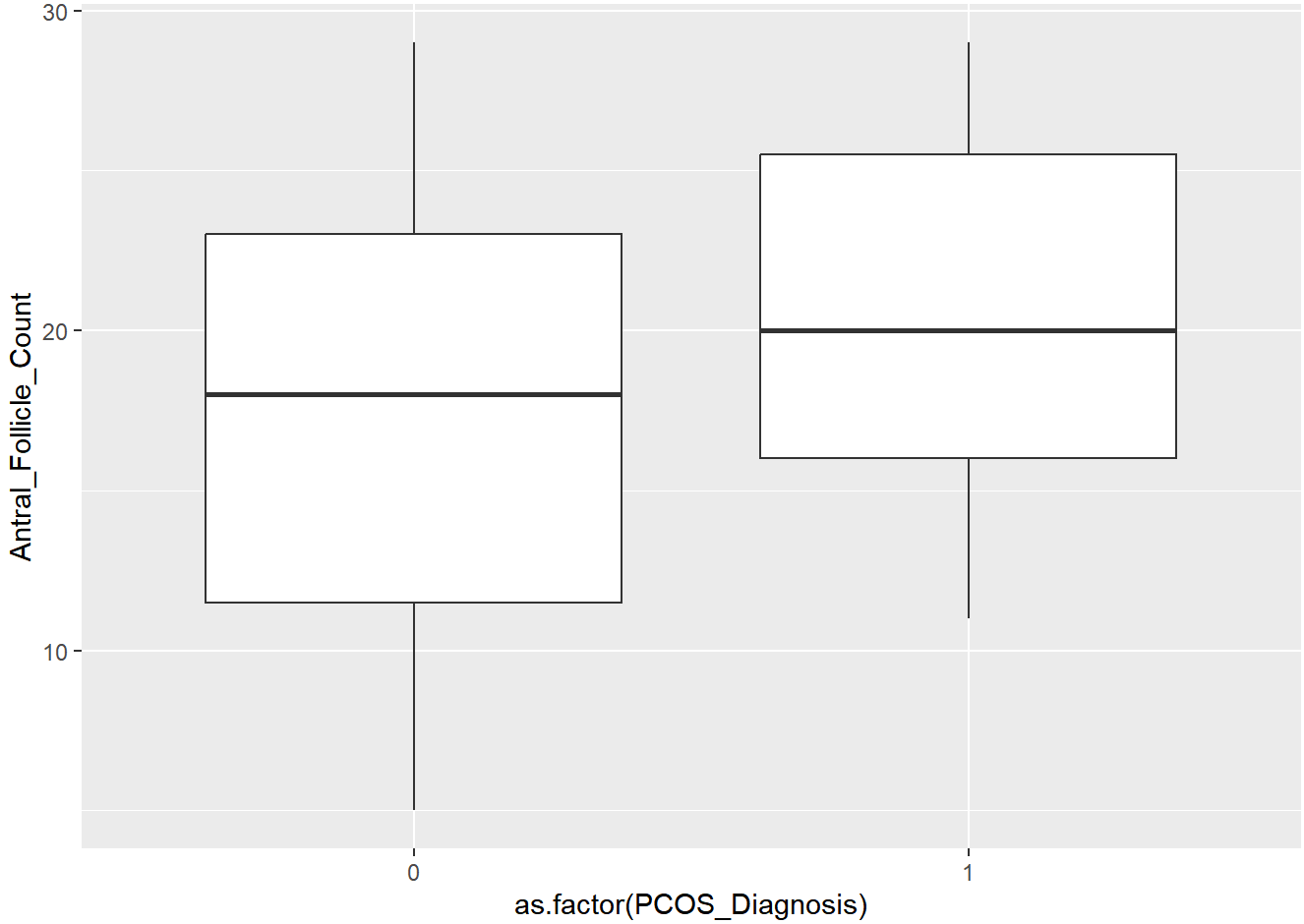

```
ggplot(pcos, aes(x = Menstrual_Irregularity, fill = as.factor(PCOS_Diagnosis))) +
  geom_bar(position = 'fill')
```




```
ggplot(pcos, aes(x= as.factor(PCOS_Diagnosis), y = Testosterone_Level.ng.dL.)) +
 geom_boxplot()
```




```
ggplot(pcos, aes(x= as.factor(PCOS_Diagnosis), y = Antral_Follicle_Count)) +
  geom_boxplot()
```




```
fit = glm(PCOS_Diagnosis ~ ., data = pcos, family = 'binomial')
summary(fit)
```

```
##
## Call:
## glm(formula = PCOS_Diagnosis ~ ., family = "binomial", data = pcos)
##
## Coefficients:
##                             Estimate Std. Error z value Pr(>|z|)
## (Intercept)                -65.55789 2656.89094  -0.025    0.980
## Age                         -0.01107    0.02293  -0.483    0.629
## BMI                          0.50589    0.06587   7.680 1.59e-14 ***
## Menstrual_Irregularity      23.33562 1328.44429   0.018    0.986
## Testosterone_Level.ng.dL.    0.05992    0.01078   5.558 2.73e-08 ***
## Antral_Follicle_Count        0.14034    0.03506   4.003 6.26e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 528.89  on 381  degrees of freedom
## Residual deviance: 168.26  on 376  degrees of freedom
## AIC: 180.26
##
## Number of Fisher Scoring iterations: 19
```

backward and forward step

```
backward_model <- stepAIC(fit, direction = "backward")
```

```
## Start:  AIC=180.26
## PCOS_Diagnosis ~ Age + BMI + Menstrual_Irregularity + Testosterone_Level.ng.dL. +
##     Antral_Follicle_Count
##
##                             Df Deviance    AIC
## - Age                        1   168.49 178.49
## <none>                           168.26 180.26
## - Antral_Follicle_Count      1   186.95 196.95
## - Testosterone_Level.ng.dL.  1   210.46 220.46
## - BMI                        1   287.31 297.31
## - Menstrual_Irregularity     1   358.31 368.31
##
## Step:  AIC=178.49
## PCOS_Diagnosis ~ BMI + Menstrual_Irregularity + Testosterone_Level.ng.dL. +
##     Antral_Follicle_Count
##
##                             Df Deviance    AIC
## <none>                           168.49 178.49
## - Antral_Follicle_Count      1   187.41 195.41
## - Testosterone_Level.ng.dL.  1   211.25 219.25
## - BMI                        1   290.90 298.90
## - Menstrual_Irregularity     1   361.02 369.02
```

```
null = glm(PCOS_Diagnosis ~ 1, data = pcos, family = 'binomial')
forward_model = stepAIC(null, scope = list(lower= null, upper = fit),direction = 'forward')
```

```
## Start:  AIC=530.89
## PCOS_Diagnosis ~ 1
##
##                              Df Deviance    AIC
## + Menstrual_Irregularity      1   344.21 348.21
## + BMI                         1   415.87 419.87
## + Testosterone_Level.ng.dL.   1   491.98 495.98
## + Antral_Follicle_Count       1   509.12 513.12
## + Age                         1   524.44 528.44
## <none>                            528.89 530.89
##
## Step:  AIC=348.21
## PCOS_Diagnosis ~ Menstrual_Irregularity
##
##                              Df Deviance    AIC
## + BMI                         1   223.60 229.60
## + Testosterone_Level.ng.dL.   1   314.44 320.44
## + Antral_Follicle_Count       1   325.95 331.95
## + Age                         1   340.01 346.01
## <none>                            344.21 348.21
##
## Step:  AIC=229.6
## PCOS_Diagnosis ~ Menstrual_Irregularity + BMI
##
##                              Df Deviance    AIC
## + Testosterone_Level.ng.dL.   1   187.41 195.41
## + Antral_Follicle_Count       1   211.25 219.25
## <none>                            223.60 229.60
## + Age                         1   222.73 230.73
##
## Step:  AIC=195.41
## PCOS_Diagnosis ~ Menstrual_Irregularity + BMI + Testosterone_Level.ng.dL.
##
##                         Df Deviance    AIC
## + Antral_Follicle_Count  1   168.49 178.49
## <none>                       187.41 195.41
## + Age                    1   186.95 196.95
##
## Step:  AIC=178.49
## PCOS_Diagnosis ~ Menstrual_Irregularity + BMI + Testosterone_Level.ng.dL. +
##     Antral_Follicle_Count
##
##        Df Deviance    AIC
## <none>      168.49 178.49
## + Age   1   168.26 180.26
```

validation-set approach

```
# training is 1/2 data
set.seed(10)
train = sample(nrow(pcos), size = nrow(pcos)/2)

fit_train = glm(PCOS_Diagnosis ~ BMI + Menstrual_Irregularity + Testosterone_Level.ng.dL.+ Antra
l_Follicle_Count, data = pcos, subset = train, family = 'binomial')

p = ifelse(predict(fit_train, newdata = pcos[-train,], type = 'response') > 0.5, 'yes', 'no')
t = table(p, pcos$PCOS_Diagnosis[-train])

(sum(diag(t))/sum(t))
```

```
## [1] 0.9057592
```

```
# training is 1/3 data
set.seed(10)
train = sample(nrow(pcos), size = nrow(pcos)/3)

fit_train = glm(PCOS_Diagnosis ~ BMI + Menstrual_Irregularity + Testosterone_Level.ng.dL. + Antr
al_Follicle_Count, data = pcos, subset = train, family = 'binomial')

p = ifelse(predict(fit_train, newdata = pcos[-train,], type = 'response') > 0.5, 'yes', 'no')
t = table(p, pcos$PCOS_Diagnosis[-train])

(sum(diag(t))/sum(t))
```

```
## [1] 0.9019608
```

```
# training is 2/3 data
set.seed(10)
train = sample(nrow(pcos), size = nrow(pcos)*(2/3))

fit_train = glm(PCOS_Diagnosis ~ BMI + Menstrual_Irregularity + Testosterone_Level.ng.dL. + Antr
al_Follicle_Count, data = pcos, subset = train, family = 'binomial')

p = ifelse(predict(fit_train, newdata = pcos[-train,], type = 'response') > 0.5, 'yes', 'no')
t = table(p, pcos$PCOS_Diagnosis[-train])

(sum(diag(t))/sum(t))
```

```
## [1] 0.9296875
```

```
# splitting data
pcos_train = pcos[train,]
pcos_test = pcos[-train,]
```

log regression

```
fit = glm(PCOS_Diagnosis ~  BMI + Menstrual_Irregularity + Testosterone_Level.ng.dL. + Antral_Fo
llicle_Count, data = pcos_train, family = 'binomial')

# accuracy
pred = predict(fit, newdata = pcos_test, type = 'response')
p = ifelse(pred > 0.5, 1, 0)
t = table(p, pcos_test$PCOS_Diagnosis)
print(t)
```

```
##
## p    0  1
##   0 59  3
##   1  6 60
```

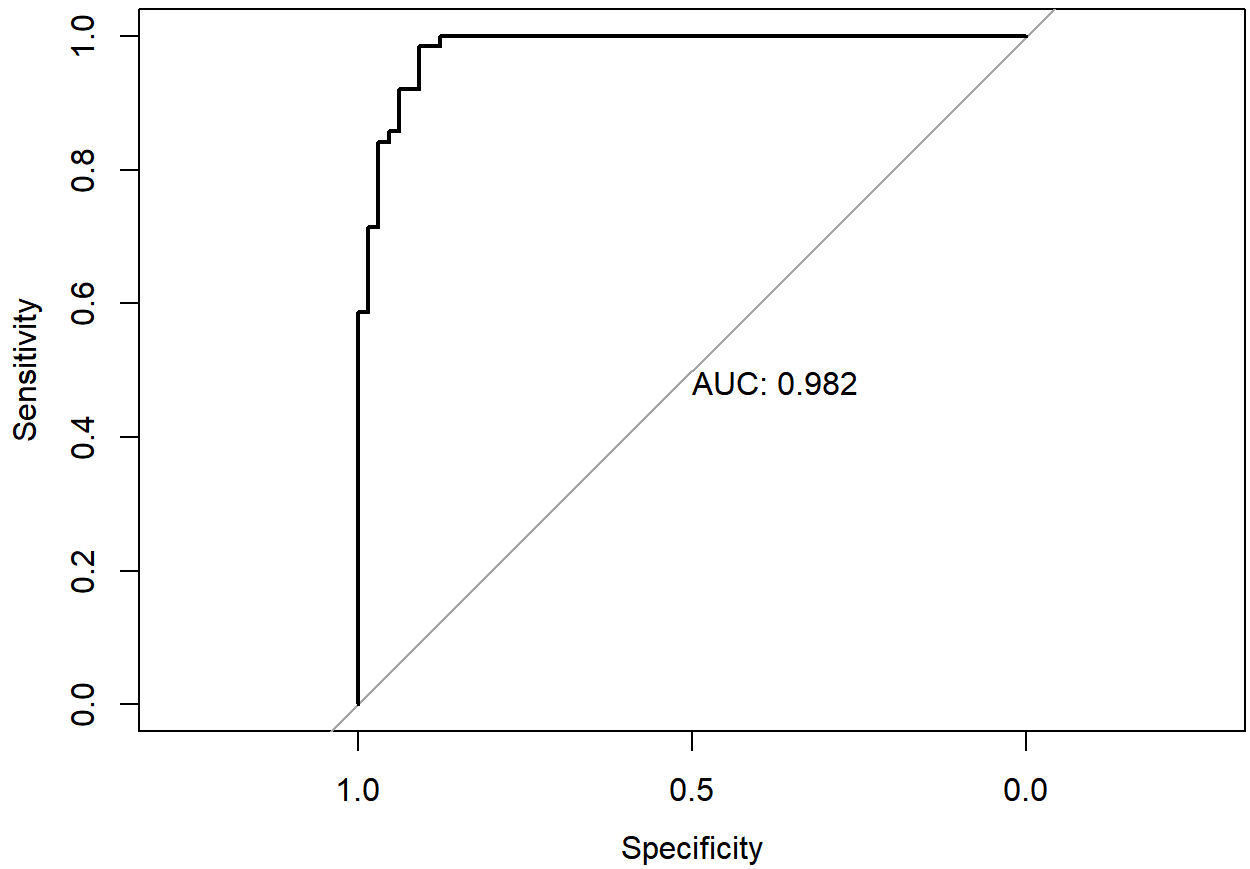
```
(fit_acc = sum(diag(t))/sum(t))
```

```
## [1] 0.9296875
```

```
# auc
plot(roc(pcos_test$PCOS_Diagnosis, pred), print.auc = TRUE)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```



LDA

```
ldafit = lda(PCOS_Diagnosis ~  BMI + Menstrual_Irregularity + Testosterone_Level.ng.dL. + Antral
_Follicle_Count, data = pcos_train)
#plot(ldafit)

# accuracy
p = predict(ldafit, newdata = pcos_test)$class
t = table(p, pcos_test$PCOS_Diagnosis)
print(t)
```

```
##
## p    0  1
##   0 52  0
##   1 13 63
```

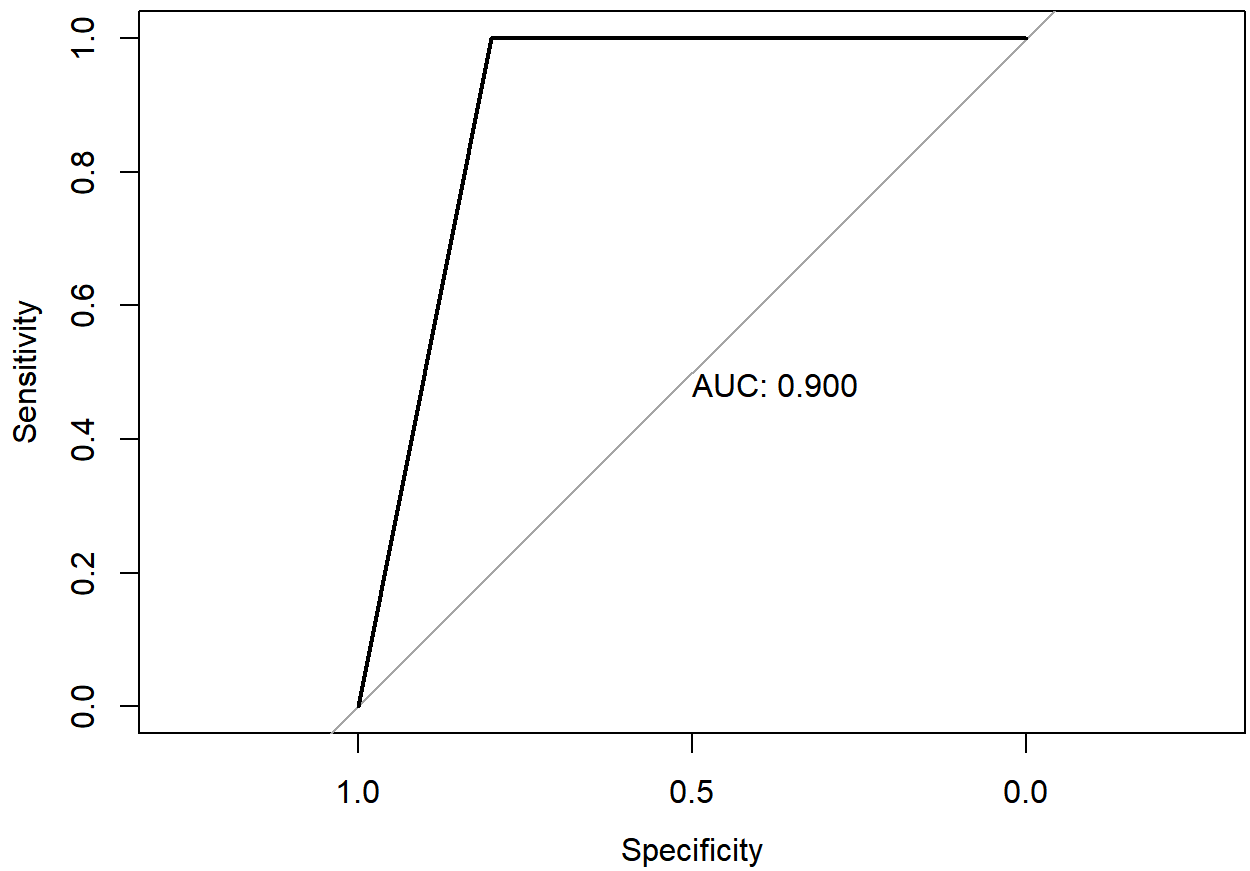
```
(lda_acc = sum(diag(t))/sum(t))
```

```
## [1] 0.8984375
```

```
# auc
pred = as.numeric(p)
plot(roc(pcos_test$PCOS_Diagnosis, pred), print.auc = TRUE)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```



QDA

```
#qdafit = qda(PCOS_Diagnosis ~ BMI + Menstrual_Irregularity + Testosterone_Level.ng.dL. + Antral
_Follicle_Count, data = pcos_train)

#acc(qdafit)

# not working!!!
```

decision tree

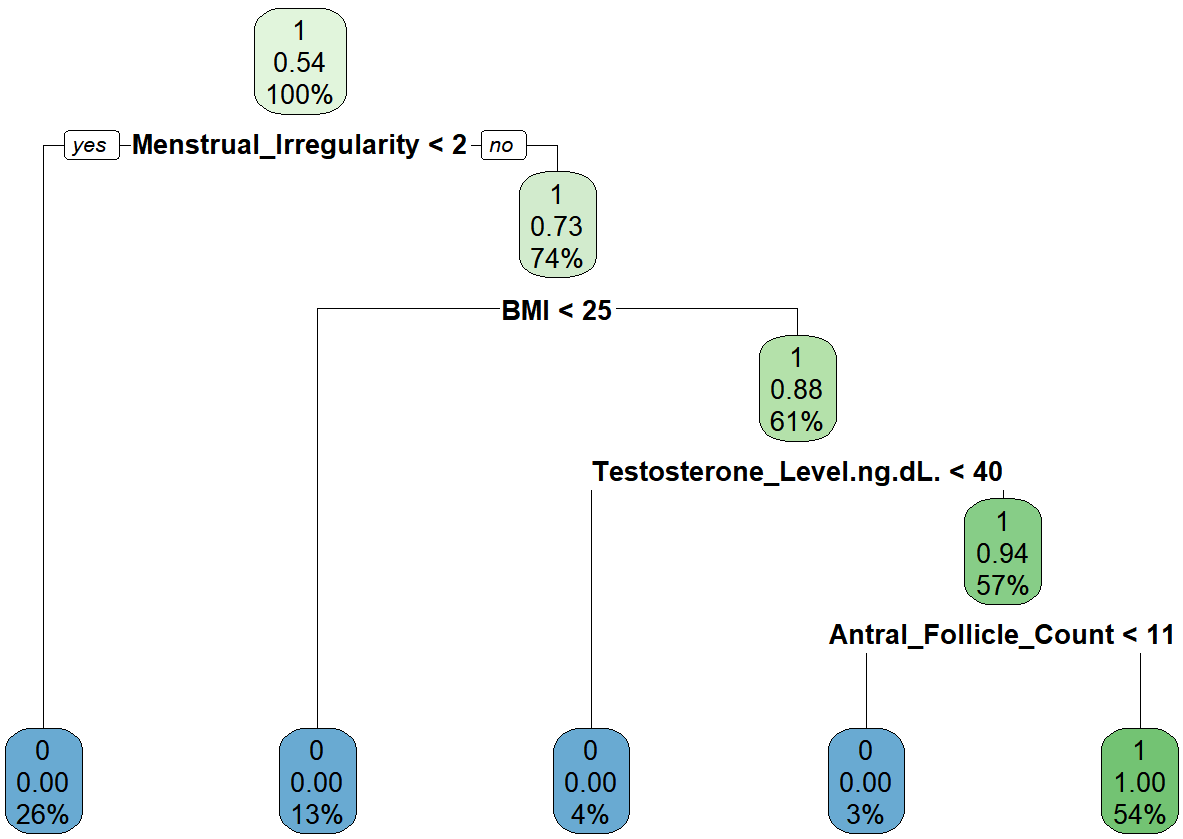
```
library(rpart.plot)
```

```
## Warning: package 'rpart.plot' was built under R version 4.5.2
```

```
## Loading required package: rpart
```

```
## Warning: package 'rpart' was built under R version 4.5.2
```

```
rtree = rpart(PCOS_Diagnosis ~ BMI + Menstrual_Irregularity + Testosterone_Level.ng.dL. + Antral
_Follicle_Count, data = pcos_train, cp = 0.008)
rpart.plot(rtree)
```

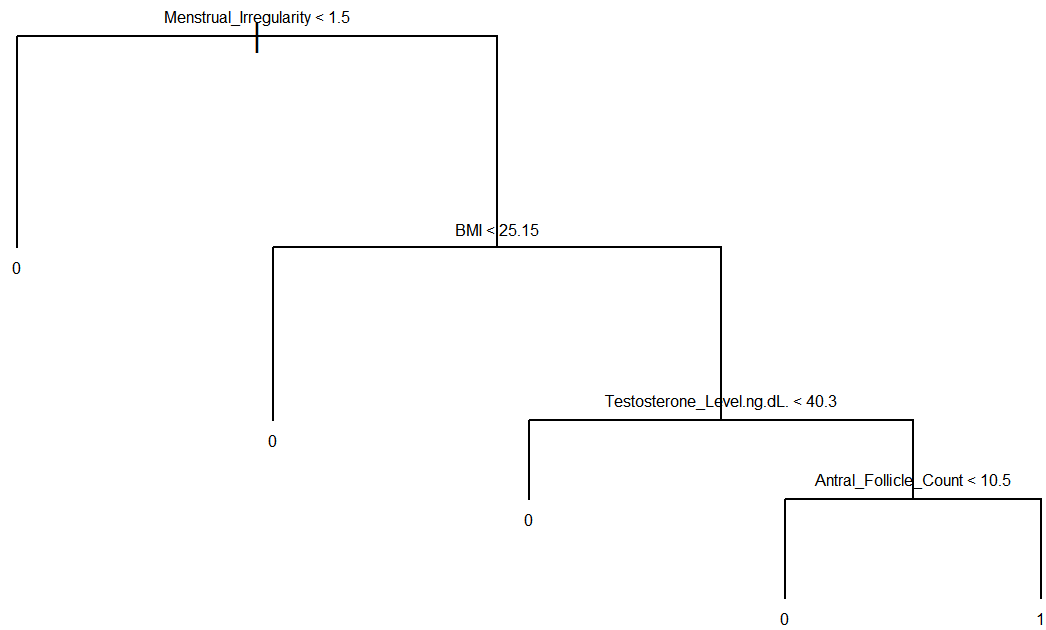



```
#text(tree, cex = 0.5, digits = 3)
```

```
library(tree)
```

```
## Warning: package 'tree' was built under R version 4.5.2
```

```
tree = tree(PCOS_Diagnosis ~ BMI + Menstrual_Irregularity + Testosterone_Level.ng.dL. + Antral_F
ollicle_Count, data = pcos_train)
plot(tree)
text(tree, cex = 0.5, digits = 3)
```




```
# accuracy
p = predict(tree, newdata = pcos_test, type = 'class')
t = table(p, pcos_test$PCOS_Diagnosis)
print(t)
```

```
##
## p    0  1
##   0 65  1
##   1  0 62
```

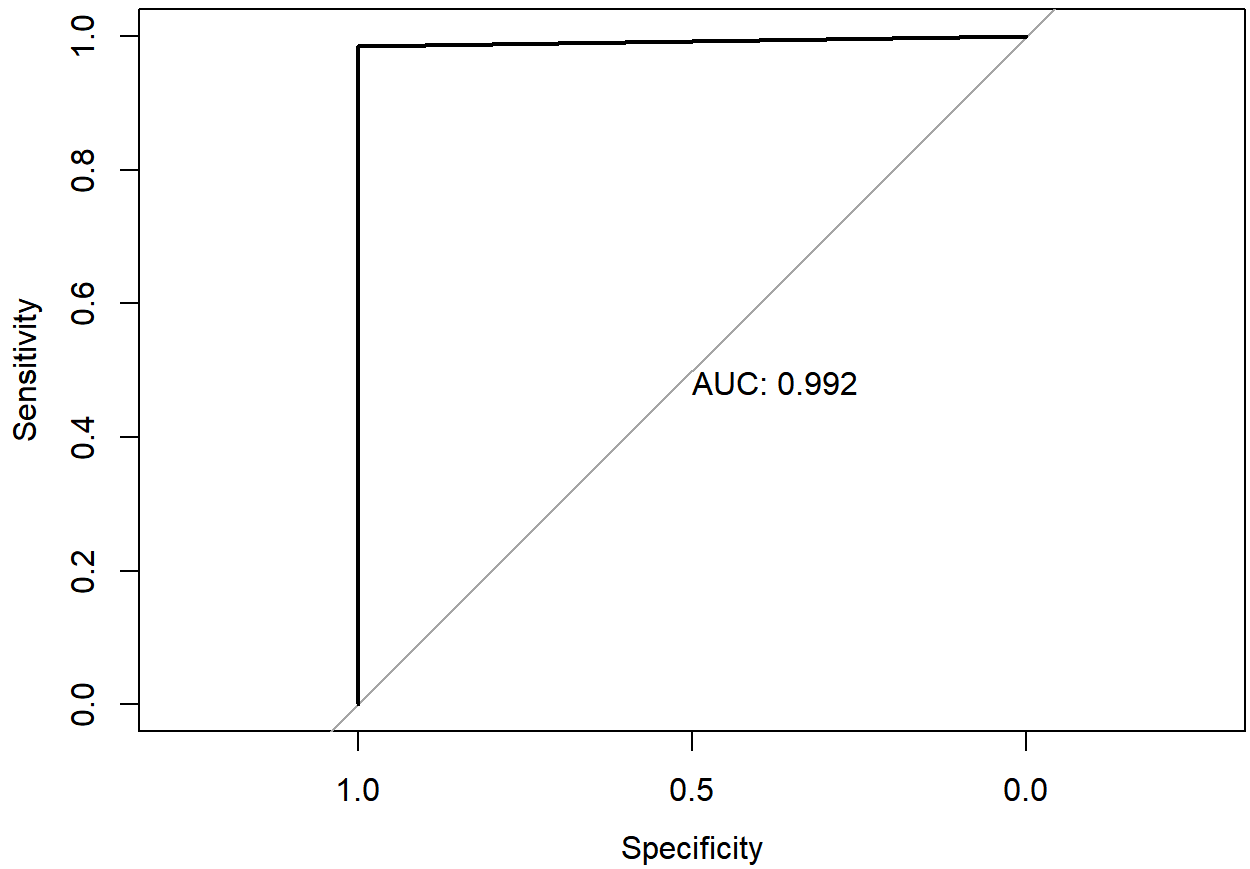
```
(tree_acc = sum(diag(t))/sum(t))
```

```
## [1] 0.9921875
```

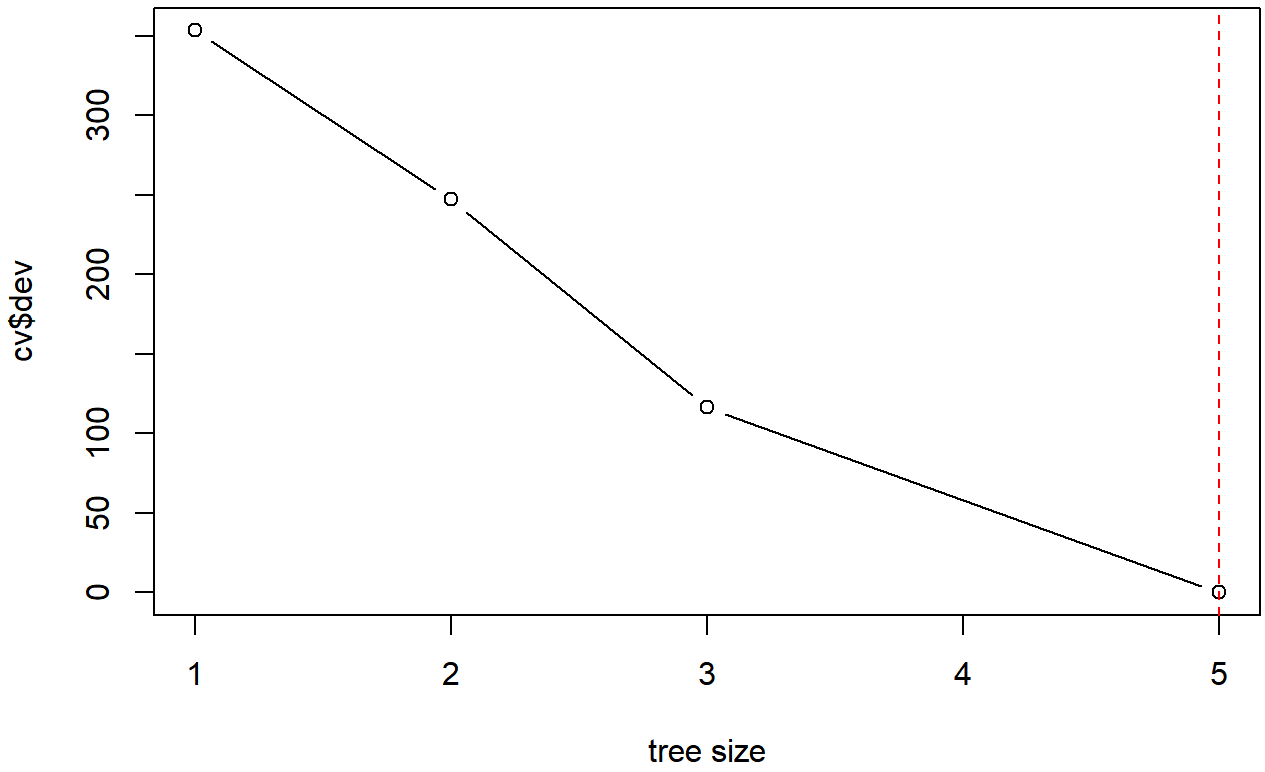
```
# auc
pred = as.numeric(p)
plot(roc(pcos_test$PCOS_Diagnosis, pred), print.auc = TRUE)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```




```
# pruned
library(tree)
cv = cv.tree(tree)

plot(cv$size, cv$dev, type = 'b', xlab = 'tree size')
min = which.min(cv$dev)
abline(v = cv$size[min], lty = 2, col = 'red')
```



random forest

```
library(randomForest)
```

```
## Warning: package 'randomForest' was built under R version 4.5.2
```

```
## randomForest 4.7-1.2
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

```
##
## Attaching package: 'randomForest'
```

```
## The following object is masked from 'package:ggplot2':
##
##     margin
```

```
## The following object is masked from 'package:dplyr':
##
##     combine
```

```
rf = randomForest(PCOS_Diagnosis ~ BMI + Menstrual_Irregularity + Testosterone_Level.ng.dL. + An
tral_Follicle_Count, data = pcos_train, mtry = 2)

# accuracy
p = predict(rf, newdata = pcos_test, type = 'class')
t = table(p, pcos_test$PCOS_Diagnosis)
print(t)
```

```
##
## p    0  1
##   0 65  1
##   1  0 62
```

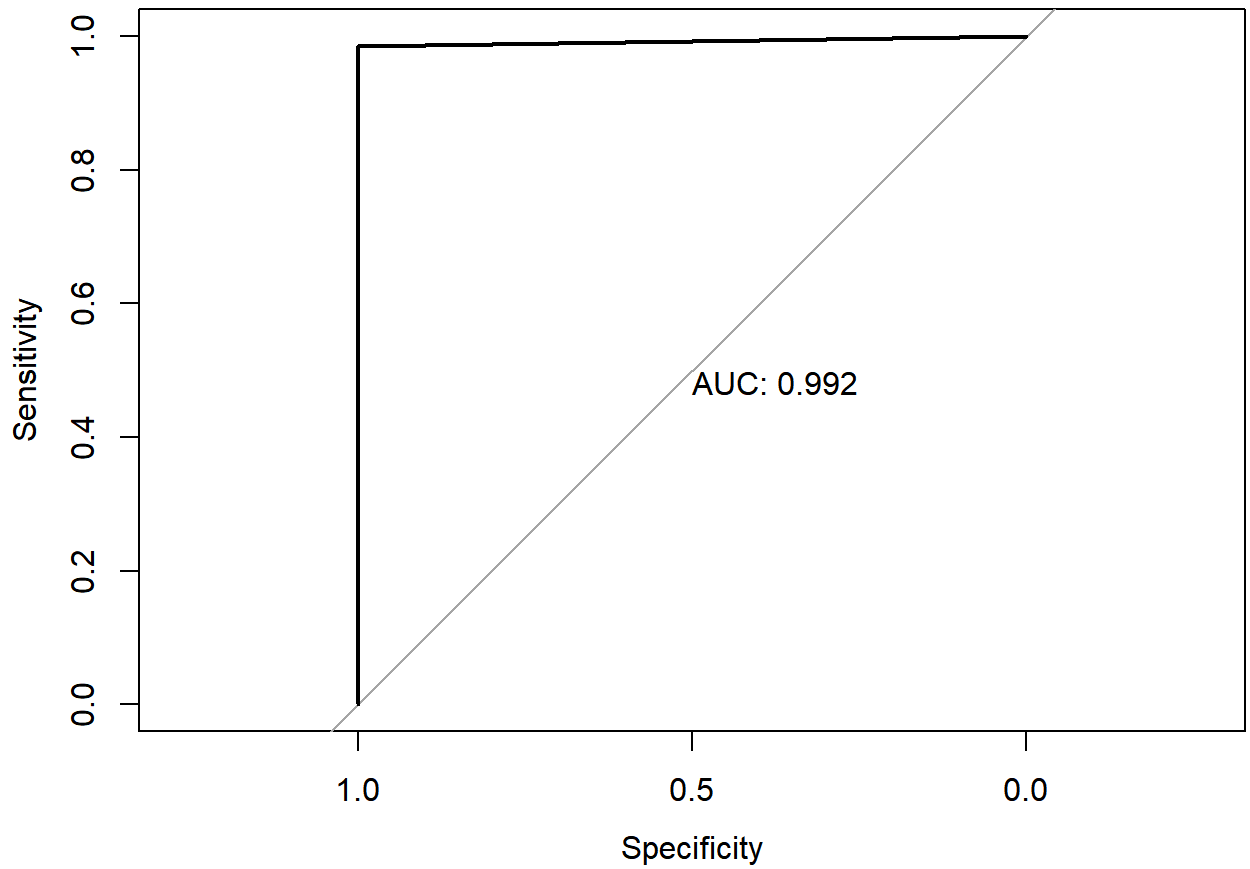
```
(rf_acc = sum(diag(t))/sum(t))
```

```
## [1] 0.9921875
```

```
# auc
pred = as.numeric(p)
plot(roc(pcos_test$PCOS_Diagnosis, pred), print.auc = TRUE)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```



bagging

```
bag = randomForest(PCOS_Diagnosis ~ BMI + Menstrual_Irregularity + Testosterone_Level.ng.dL. + A
ntral_Follicle_Count, data = pcos_train, mtry = 4)

p = predict(bag, newdata = pcos_test, type = 'class')
t = table(p, pcos_test$PCOS_Diagnosis)
print(t)
```

```
##
## p    0  1
##   0 65  1
##   1  0 62
```

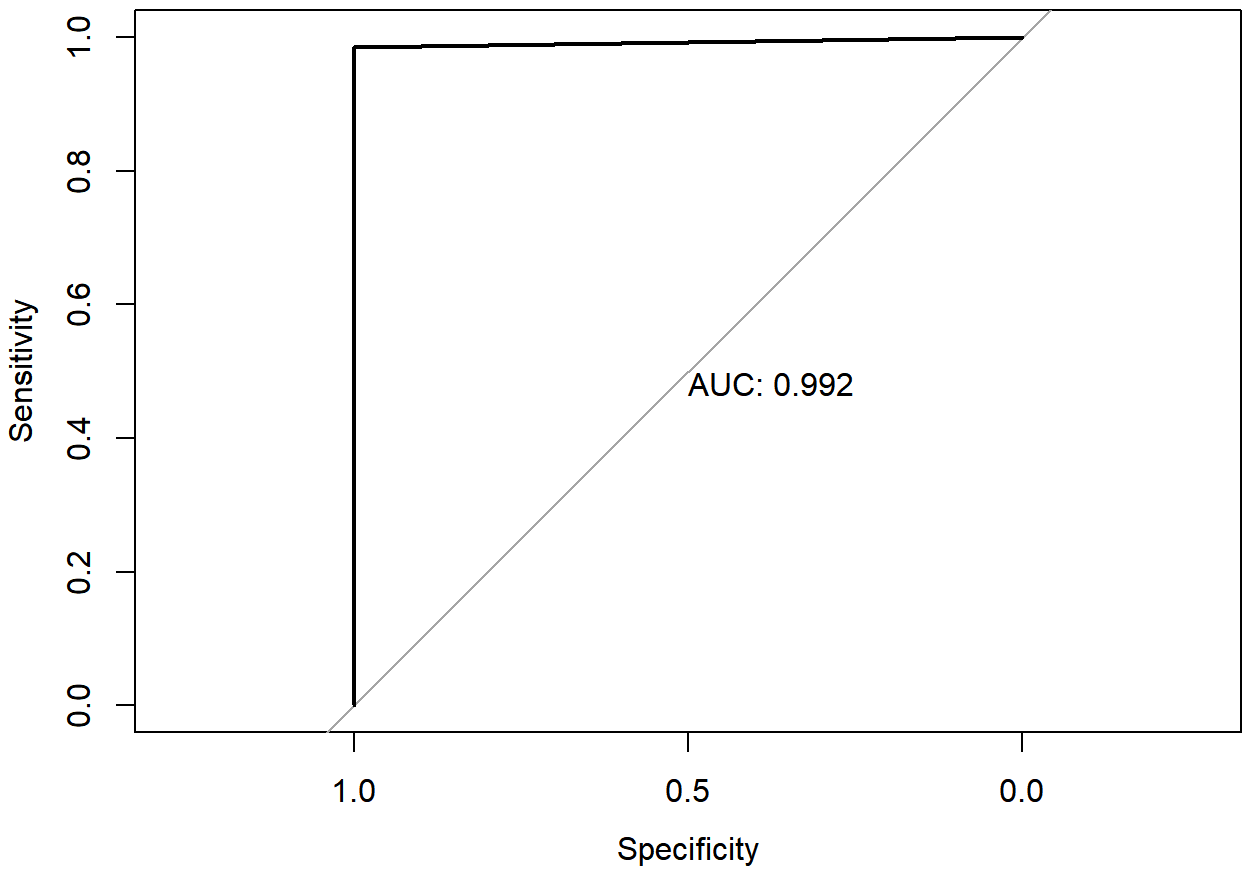
```
(bag_acc = sum(diag(t))/sum(t))
```

```
## [1] 0.9921875
```

```
# auc
pred = as.numeric(p)
plot(roc(pcos_test$PCOS_Diagnosis, pred), print.auc = TRUE)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```



naive bayes

```
library(e1071)
```

```
##
## Attaching package: 'e1071'
```

```
## The following object is masked from 'package:ggplot2':
##
##     element
```

```
library()
nbfit = naiveBayes(PCOS_Diagnosis ~ BMI + Menstrual_Irregularity + Testosterone_Level.ng.dL. + A
ntral_Follicle_Count, data = pcos_train)

p = predict(nbfit, newdata = pcos_test, type = 'class')
t = table(p, pcos_test$PCOS_Diagnosis)
print(t)
```

```
##
## p    0  1
##   0 35  0
##   1 30 63
```

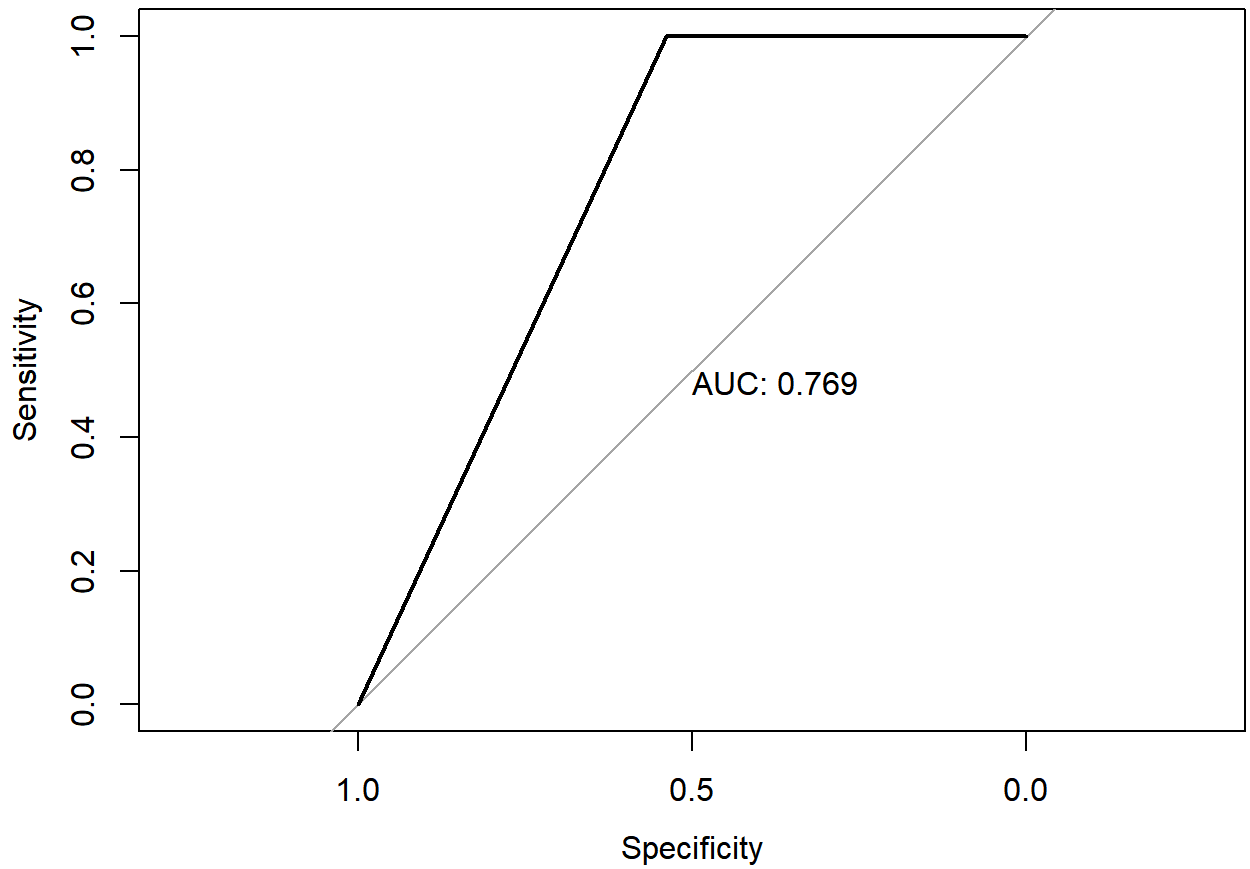
```
(nb_acc = sum(diag(t))/sum(t))
```

```
## [1] 0.765625
```

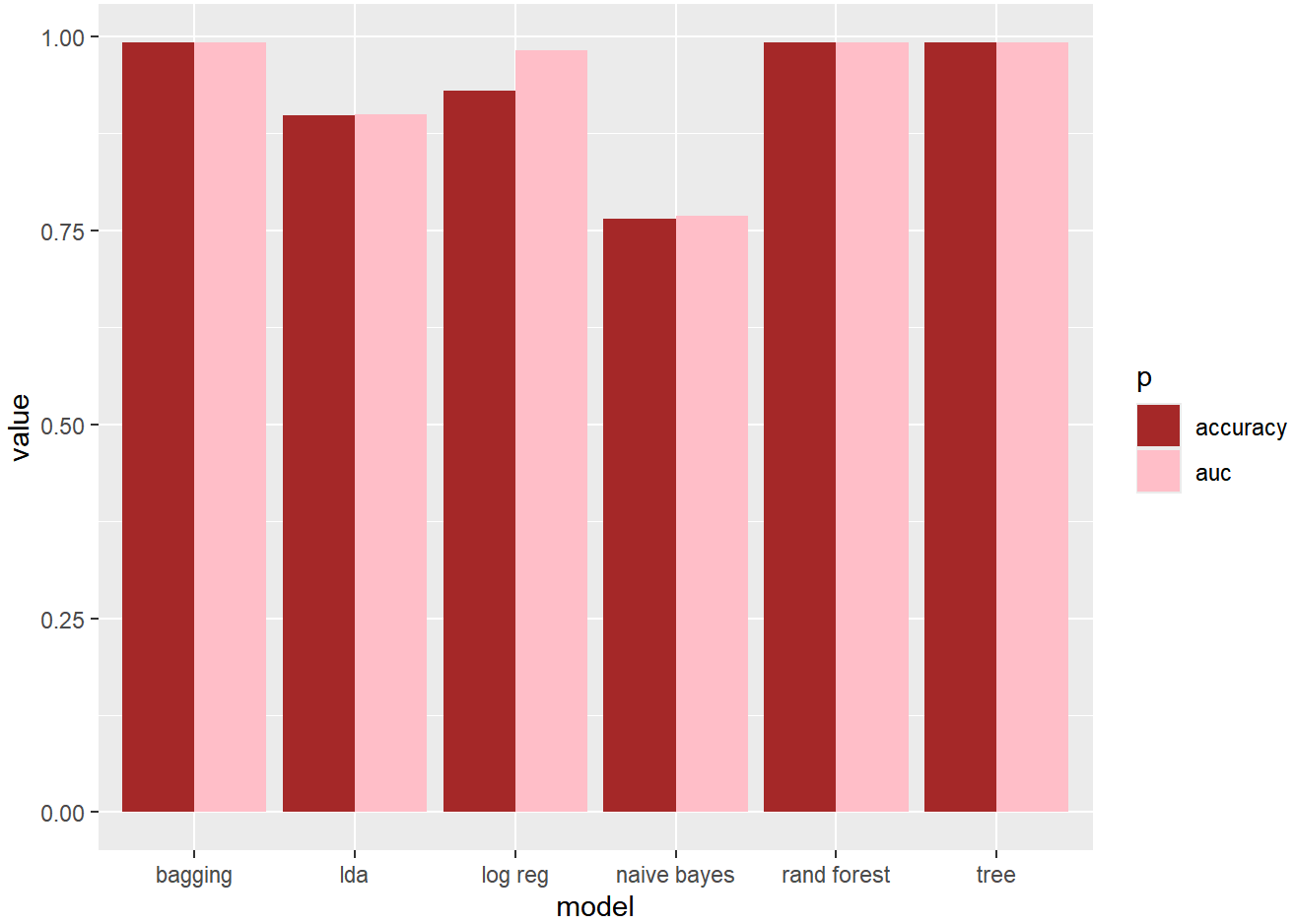
```
# auc
pred = as.numeric(p)
plot(roc(pcos_test$PCOS_Diagnosis, pred), print.auc = TRUE)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```




```
library(tidyr)
table = data.frame(model = c('log reg', 'lda', 'tree', 'rand forest', 'bagging', 'naive bayes'),
                     accuracy = c(fit_acc, lda_acc, tree_acc, rf_acc, bag_acc, nb_acc),
                     auc = c(0.982, 0.9, 0.992, 0.992, 0.992, 0.769))
table2 = table |>
  gather('p', 'value', -model)

ggplot(table2) + geom_col(aes(x = model, y = value, fill = p), position = 'dodge') + scale_fill_
manual(values = c('brown','pink'))
```


1.00
0.75
0.50
0.25
0.00
value
bagging
lda
log reg
naive bayes
rand forest
tree
model
p
accuracy
auc